# Accelerating Agricultural Research Through Interoperable Genetic and Phenotypic Variation Data

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## Making Genetic and Phenotypic Variation Data FAIR for Agricultural Research

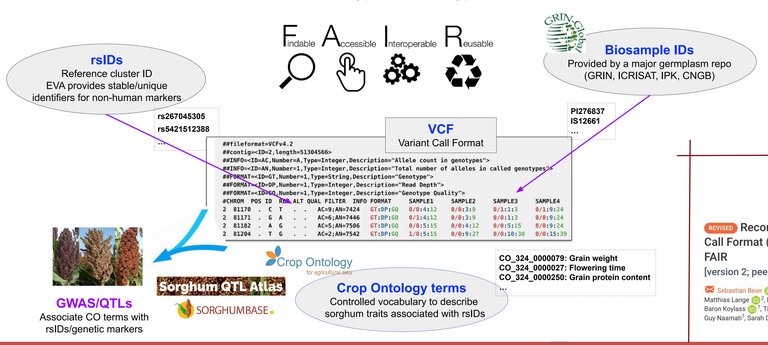
**Summary**

Agriculture is a key component of the One Health Joint Plan of Action (OH JPA), a global initiative focused on developing sustainable solutions to address health threats. Our work supports the OH JPA pathway of data, evidence, information systems, and knowledge exchange. FAIR data standards are crucial for integrating genetic and phenotypic datasets, enabling comparisons, and providing broader insights. While these standards are well-established in biomedical research, they've only recently been adopted in agriculture.

Historically, agricultural data sharing relied on personal networks, requiring significant effort to clean and harmonize data, limiting collaboration. To improve this, we worked with the AgBioData Standards for Genetic Variation Working Group to evaluate practices and review data formats. We introduced rsIDs for genetic markers in the SorghumBase and Gramene databases, ensuring consistency across breeding lines and landraces, and partnered with germplasm repositories to implement standard biosample identifiers, thus enhancing interoperability across Ag data resources.

Our efforts show that data standards not only improve data sharing but also support re-analysis, and boost research visibility, collaboration, and impact.

### Standards for Genotypic & Phenotypic Variation




### How to make agricultural data FAIR?

- **Use standard formats**
  - VCF for genotypic data (naturally occurring SNPs & EMS-induced mutations)
  - Interoperability between VCF and other formats, such as HapMap
  - Ontology template for phenotypic data (e.g., DeltaBreed, MIAAPE)
- **Supply complete metadata** (germplasm ID, geolocation, method, units)
- **Embrace ontologies** (hierarchically organized controlled vocabularies to describe phenotypes - plant PO, trait TO, crop CO, experimental factor EFO)
- **Use persistent and unambiguous identifiers**
  - rsIDs for SNPs (and EMS mutations)
  - Germplasm identifiers given by major repositories like GRIN (PI#), ICRISAT (IS#)
- **Put your data in a FAIR long-term stable repository**
  - Genome assembly → INSDC
  - SNPs → European Variation Archive (EVA)
  - Traits → Community DB (e.g., Breeding Insight - specialty crops)
- **Cite, share freely and encourage others**

### Background: review of the literature and existing Ag resources

REVISED **Recommendations for the formatting of Variant Call Format (VCF) files to make plant genotyping data FAIR**
[version 2; peer review: 2 approved]

Sebastian Beier[1,2], Anne Fiebig[1,2], Cyril Pommier[1], Isuru Liyanage[1], Matthias Lange[1,2], Paul J. Kersey[3], Stephan Weise[1], Richard Finkers[4,5], Baron Kaylass[1], Timothee Cezard[3], Melanie Courtot[3,6], Bruno Contreras-Moreira[1], Guy Naamati[3], Sarah Dyer[3], Uwe Scholz[1]

elixir

FAANG
Functional Annotation of ANImal Genomes (FAANG) Project
— A coordinated international action to accelerate Genome to Phenome

Recommended use of rsIDs in VCF and major germplasm identifiers in metadata

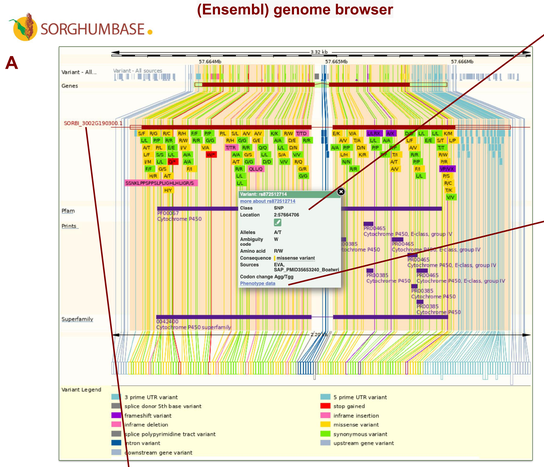
## Genotypic & Phenotypic Variation in Gramene & PanGenome Sites

### Variant identifiers replaced with standard rsIDs in Gramene PanGenomes

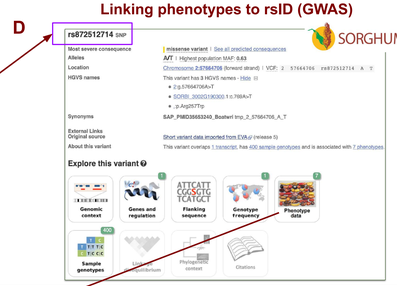
| Gramene PanGenome | Reference Crop (release #) | # SNPs (M) | # rsIDs (M) |
|---|---|---|---|
| SORGHUMBASE — Genomic resources for the sorghum research community | Sorghum (R9) | 78 | 41 |
| MAIZE Gramene — Comparative plant genomics focused on maize varieties | Maize (R5) | 220 | 79 |
| OryzaGene — Comparative plant genomics focused on rice varieties | Rice (R8) | 66 | 27 |
| GRAPEVINE Gramene — Comparative plant genomics focused on grapevine varieties | Grape (R4) | 0.5 | 0.3 |

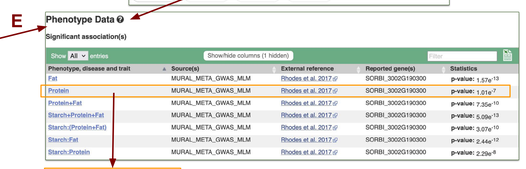
Below are visual displays of genotypic and phenotypic data from the SorghumBase genome browser: **A)** Genetic variants along the SORBI_3002G190300 gene (Cytochrome P450) are color-coded by their predicted functional consequences and mapped to protein domains. rs872512714, a missense variant, is shown in the inset. Click on a variant for more details, including links to the individual variant page (D) and associated phenotypes (E). **B)** The gene is associated with multiple phenotypes, represented by QTLs from the OZ Sorghum QTL Atlas (Mace et al, 2018), such as protein content. **C)** A table of all sorghum QTLs associated with protein content. **D)** Variant page for rs872512714. Click on the icons for specific information, including seven associated traits. **E)** Traits associated with rs872512714, as identified in the meta-analysis by Mural et al (2021; *10.1093/genetics/iyab087*). Click on a trait to view all variant loci associated with it. **F)** A list of all sorghum short variants (SNPs and indels) linked to protein content.

### Visualizing genotypic and phenotypic variation data in the SorghumBase (Ensembl) genome browser

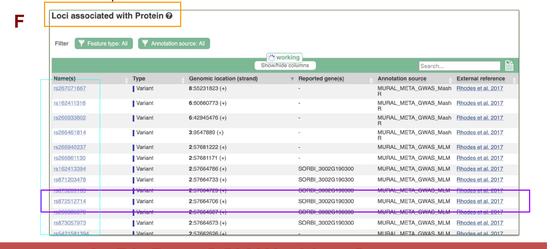


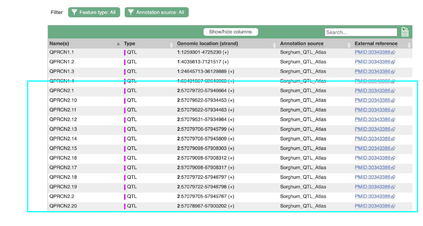

### Linking phenotypes to rsID (GWAS)




### Linking phenotypes to genomic loci (QTLs)




### SorghumBase

- GWAS link rsID to traits & traits to rsIDs
- QTLs link genomic regions to traits

### Future Plans for Gramene PanGenomes

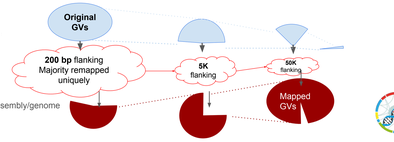
- Identify GWAS studies
- Identify rsID delimiting QTLs

## Adoption of rsIDs in Gramene's Pan-Genomes

### Workflow

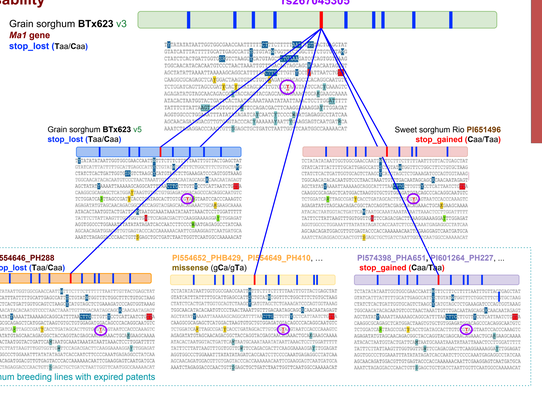
SORGHUMBASE

- Map 49 million EVA R5 rsIDs across different versions of the reference genome and to pan-genomes
- Benchmark on sorghum genomes
  - ~99% mapping rate from sorghum reference assembly v3 to v5
  - ~87% mapping rate (41 million) across sorghum pan-genome




https://github.com/EBIvariation/variant-remapping/blob/master/README.md

### rsIDs provide stable identifiers for traversing Pan-Genomes improving usability




Flanking sequence for variant rs267045305 in the SorghumBase genome browser. The target variant is highlighted in red inside a purple circle, and neighboring variants are marked with highlighted letters and ambiguity codes. It maps to *Ma1*, a gene that affects how sensitive flowering time is to the length of daylight. The sorghum BTx623 reference genome is homozygous for the minor allele T; the alternative allele C is predicted to result in the "loss" of a stop codon (relative to the reference) rendering a longer Ma1 protein. This protein-truncating variant was previously reported as a nonsense mutation (Murphy et al., 2011; Klein et al, 2015; Grant et al., 2023). Mapping this rsID to other sorghum genomes predicted a missense mutation (Ala → Val) in several breeding lines, indicating an upstream nucleotide insertion or a sequencing artifact. Moreover, this region seemed to be duplicated in PI562621_PH538 (not shown).

## Biocurating Germplasm, Markers, Traits

### Manual biocuration is needed to standardize sorghum germplasm, genetic markers, and traits

Following the biocuration of sorghum germplasm, metadata were gathered for indexing, allowing individual accessions to be ordered from major germplasm repositories, such as GRIN Global, via the Germplasm tab of the SorghumBase search interface. Marker biocuration was performed semi-automatically with manual checks. Trait biocuration was done using the Trait (TO) and Sorghum Crop (CO_324) ontologies.

#### Germplasm Biocuration Examples

| GRIN: PI# | cultivar name | IS | SAP | SC | inventory | other | alt. spellings |
|---|---|---|---|---|---|---|---|
| PI 616029 | B35 | - | SAP-37 | - | NSSL 46071S | - | - |
| PI 533752 | BTx642 / S4.K.04 (Wildwood res.) | IS24030 | SAP-127 | SC103* / SC103-14E* | NSSL 339139 | - | - |
| PI 580681 | Rio | - | - | - | - | SI-179 | RIO, 'RIO' |
| PI 565795 | | IS 9606 | - | - | - | - | |
| PI 651496 | | - | SAP-105 | - | NSL 40230 | MBR 55-1 | |
| PI 642082 | | - | SAP-55 | - | Grif 15995 MN 2609 SA 5257-5 | Standard Early Hegari | |
| PI 586641 | Early Hegari | IS 27816 | - | - | Grif 8862 | 3B/106-11 | Early_Hegari, EarlyHegari, Early hegari, EARLY HEGARI, 'EARLY HEGARI' |
| PI 573372 | | IS 9649 | - | - | - | SU 502 | |
| PI 569715 | | IS 8662 | - | - | NSL 56267 | E 516 | |
| PI 655029 | | IS 107 | - | - | NSL 4009 | MN 137 | |
| PI 290212 | | - | - | - | - | MN 6496 | |
| - | | - | - | - | - | - | - |
| - | IBC/E-38432 | - | - | - | - | - | IBC_E38432, 38432 |

#### Genetic Marker Biocuration Examples

| species | rsID# | marker name1 | marker name2 | other names | curent assembly coordinates | alternative assembly coordinates |
|---|---|---|---|---|---|---|
| sorghum | rs267045305 | tmp_6_40314507_T_C | 6:41649372_T_C,SGT_Panel | - | 6:40314507 | 6:41669372 |
| sorghum | rs872512714 | tmp_2_57664706_A_T | 2:58937711:A_T:SGT_Panel | - | 2:57664706 | 2:58837771 |
| rice | rs1912995?? | vcZAA3J9I | TBG02574 | id500957&, 1052235134 | 1:22355067 | 1:22292508 |
| rice | rs1912995?? | vcZAA3J9H | TBG02575 | - | 1:22356238 | 1:22299679 |

#### Trait Biocuration Examples

| Breeder1 Trait Name | Breeder2 Trait Name | Trait Ontology Term | Trait Ontology Trait Name | Sorghum Crop Ontology Term | Soghum Crop Trait Name |
|---|---|---|---|---|---|
| Anthesis (Days to Anthesis) | Flowering (DTF, Days to Flowering) | TO:0000246 | flowering time trait | CO_324:0000027 | Flowering time |
| Grain color | Pericarp color | TO:0000287 | pericarp color | CO_324:0000045 | Grain color |
| Panicle compactness | Panicle spread (compact/open) | - | - | CO_324:0000358 | Panicle compactness and shape |

Germplasm Biocuration Examples

Germplasm metadata is needed to connect to a major repository such as GRIN or IRRI

Gene Search: Germplasm Tab

Sample germplasm page (Passport Tab) in GRIN

Sorghum genes affected with a putative truncating variant (LOF) in a given sorghum accession

### FAIR Standards Aid Data Interpretation & Identification of Potential Artifacts

A graphical representation of a multiple protein alignment for the *Ma1* phylogenetic tree, showing the predicted gain (full-length) and loss-of-function (truncated) gene products across the sorghum pan-genome due to the presence of the reference or alternative allele for rs267045305 (relative position shown with a purple arrow). The tree was constructed using canonical transcripts, following the Ensembl Compara pipeline. This visualization aids in identifying genes for structural biocuration, helping to refine gene models and alternative transcripts.

Gene models for *Ma1* (stop lost) in v3 vs split genes in v5). The *Ma1* gene co-localizes with QTLs related to traits such as days to flowering, plant height, and grain weight, which is consistent with *Ma1* being a maturity gene.

## AgBioData SGV Working Group

### Replacing temporary SNP identifiers with standard rsIDs

The sorghum community developed a marker panel with AgriPlex:

- rsIDs assigned for 2.4K sorghum SNPs from EVA R5
- 26 markers without rsIDs were assigned one by the EVA's fast track
- rsID list and methodology shared with companies handling similar marker panels

SORGHUMBASE · AgriPlex GENOMICS · Genetic Innovation TOOLBOX · European Variation Archive · DArT diversity arrays technology · intertek Total Quality. Assured.

### Working with commercial providers to support adoption of rsIDs

Commercial genotyping service providers

Seed companies

Supported by USDA-ARS 8062-21000-051-000D